

ESSENTIAL OF DATA SCIENCE ASSIGNMENT 2

DATA ANALYSIS

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Twitter US Airline Sentiment Dataset

1. Dataset Description

The dataset includes tweet ID, airline name, sentiment, sentiment confidence, negative reason, tweet text, timestamp, and user timezone.

2. Tools Used

- Python
- NumPy
- Pandas

3. Setup Code

```
import numpy as np
import pandas as pd

df = pd.read_csv("Tweets.csv")
df['tweet_created'] = pd.to_datetime(df['tweet_created'])
```

DATA LOADED

```
print(df.head())
```

```
      tweet_id airline_sentiment ... tweet_location
user_timezone
0  570306133677760513      neutral ...      NaN Eastern
Time (US & Canada)
1  570301130888122368    positive ...      NaN Pacific
Time (US & Canada)
2  570301083672813571      neutral ...  Lets Play Central
Time (US & Canada)
3  570301031407624196    negative ...      NaN Pacific
Time (US & Canada)
4  570300817074462722    negative ...      NaN Pacific
Time (US & Canada)

[5 rows x 15 columns]
```

4. Problem Statements, Solutions and Analysis

Problem 1: Analyze sentiment distribution

Explanation:

This analyzes how customer sentiments are distributed across the dataset. It helps understand overall satisfaction levels.

```
print("\n1 Sentiment Distribution")
print(df['airline_sentiment'].value_counts())
```

OUTPUT

```
1 Sentiment Distribution
airline_sentiment
negative      9178
neutral      3099
positive      2363
Name: count, dtype: int64
```

Interpretation:

More negative tweets indicate dissatisfaction.

Problem 2: Airline with highest negative sentiment

Explanation:

This identifies which airline has the most negative feedback from customers.

```
print("\n2 Airline with Highest Negative Sentiment")
neg = df[df['airline_sentiment'] == 'negative'].groupby('airline').size()
print(neg.idxmax())
```

Output

```
2 Airline with Highest Negative Sentiment
United
```

Problem 3: Airline with highest positive sentiment

Explanation:

This finds the airline with the highest number of positive tweets.

```
print("\n3 Airline with Highest Positive Sentiment")
pos = df[df['airline_sentiment'] == 'positive'].groupby('airline').size()
print(pos.idxmax())
```

Output

```
3 Airline with Highest Positive Sentiment
Southwest
```

Interpretation:

Indicates best-performing airline.

Problem 4: Sentiment percentage

Explanation:

This calculates percentage share of each sentiment category.

```
print("\n4 Sentiment Percentage")
print(df['airline_sentiment'].value_counts(normalize=True) * 100)
```

Output

```
4 Sentiment Percentage
airline_sentiment
negative    62.691257
neutral    21.168033
positive    16.140710
Name: proportion, dtype: float64
```

Interpretation:

Helps compare sentiment proportion.

Problem 5: Sentiment trend over time

Explanation:

This analyzes how sentiment changes over time.

```
print("\n5 Sentiment Trend (Sample)")
trend =
df.groupby(df['tweet_created'].dt.date)['airline_sentiment'].value_counts()
print(trend.head(10))
```

Output

```
tweet_created  airline_sentiment
2015-02-16     negative           3
                neutral           1
2015-02-17     negative        838
                neutral        297
                positive        273
2015-02-18     negative        736
                neutral        335
                positive        273
2015-02-19     negative        751
                neutral        329
Name: count, dtype: int64
```

Interpretation:

Shows spikes in complaints.

Problem 6: Most common negative reason

Explanation:

This identifies the most frequent reason behind negative tweets.

```
print("\n6 Most Common Negative Reason")
print(df['negativereason'].value_counts().idxmax())
```

Output:

```
6 Most Common Negative Reason
Customer Service Issue
```

Interpretation:

Highlights main issue.

Problem 7: Airline with most delay complaints

Explanation:

This focuses on delay-related complaints.

```
print("\n7 Airline with Most Delay Complaints")
delay = df[df['negativereason'] == 'Late Flight'].groupby('airline').size()

if not delay.empty:
    print(delay.idxmax())
else:
    print("No data")
```

Output:

```
7 Airline with Most Delay Complaints
United
```

Interpretation:

Shows airline with delays.

Problem 8: Sentiment per airline

Explanation:

This compares sentiment distribution across airlines.

```
print("\n8 Sentiment per Airline")
print(df.groupby('airline')['airline_sentiment'].value_counts())
```

Output:

```
8 Sentiment per Airline
airline      airline_sentiment
American    negative          1960
             neutral           463
             positive          336
Delta        negative          955
             neutral           723
             positive          544
Southwest    negative         1186
             neutral           664
             positive          570
US Airways   negative         2263
             neutral           381
             positive          269
United       negative         2633
             neutral           697
             positive          492
Virgin America negative         181
             neutral           171
             positive          152
Name: count, dtype: int64
```


Interpretation:

Helps compare airline performance.

Problem 9: Average sentiment confidence

Explanation:

This calculates average confidence of sentiment classification.

```
print("\n9 Average Sentiment Confidence")
print(df.groupby('airline')['airline_sentiment_confidence'].mean())
```

Output:

```
9 Average Sentiment Confidence
airline
American      0.917352
Delta         0.869878
Southwest     0.886516
US Airways    0.921578
United        0.900878
Virgin America 0.876086
Name: airline_sentiment_confidence, dtype: float64
```

Interpretation:

Higher values indicate reliability.

Problem 10: Low confidence tweets

Explanation:

This filters tweets with low sentiment confidence.

```
print("\n10 Low Confidence Tweets")
print(df[df['airline_sentiment_confidence'] < 0.5].head())
```

Output:

```
10 Low Confidence Tweets
      tweet_id airline_sentiment ... tweet_location
user_timezone
1    570301130888122368      positive ...      NaN Pacific
Time (US & Canada)
114  569861209781989377      positive ...      Earth Eastern
Time (US & Canada)
142  569615109363929088      neutral ... New York City Eastern
Time (US & Canada)
183  569318130356502528      positive ...      Worldwide
Caracas
309  568800805475782656      neutral ... New York City Eastern
Time (US & Canada)

[5 rows x 15 columns]
```

Interpretation:

Shows uncertain data.

Problem 11: Peak tweeting hours

Explanation:

This finds the time when most tweets are posted.

```
df['hour'] = df['tweet_created'].dt.hour
print("\n11 Peak Tweet Hour")
print(df['hour'].value_counts().idxmax())
```

Output:

```
11 Peak Tweet Hour
9
```

Interpretation:

Shows peak activity time.

Problem 12: Tweets per timezone

Explanation:

This analyzes tweet distribution across timezones.

```
print("\n12 Top Timezones")
print(df['user_timezone'].value_counts().head())
```

Output:

```
12 Top Timezones
user_timezone
Eastern Time (US & Canada)    3744
Central Time (US & Canada)    1931
Pacific Time (US & Canada)    1208
Quito                          738
Atlantic Time (Canada)        497
Name: count, dtype: int64
```

Interpretation:

Shows most active regions.

Problem 13: Word count analysis

Explanation:

This calculates number of words in each tweet.

```
df['word_count'] = df['text'].astype(str).apply(lambda x: len(x.split()))
print("\n13 Average Word Count")
print(np.mean(df['word_count']))
```

Output:

```
13 Average Word Count
17.65341530054645
```

Interpretation:

Shows average tweet length.

Problem 14: Longest tweets

Explanation:

This identifies tweets with highest word count.

```
print("\n14 Longest Tweets")
print(df.sort_values(by='word_count', ascending=False).head())
```

Output:

```
14 Longest Tweets
      tweet_id  airline_sentiment  ...  hour  word_count
8062  568823603627565056          negative  ...    9         36
9435  569938939122991104          negative  ...   11         33
1189  569901512190767105          negative  ...    8         32
14295 569641259826061312          negative  ...   15         32
 537   570290993339244547          negative  ...   10         32

[5 rows x 17 columns]
```

Interpretation:

Long tweets indicate detailed feedback.

Problem 15: Word count vs sentiment

Explanation:

This compares tweet length across sentiments.

```
print("\n15 Word Count vs Sentiment")
print(df.groupby('airline_sentiment')['word_count'].mean())
```

Output:

```
15 Word Count vs Sentiment
airline_sentiment
negative      19.686206
neutral      14.408842
positive      14.013119
Name: word_count, dtype: float64
```

Interpretation:

Negative tweets are usually longer.

Problem 16: Highly negative tweets

Explanation:

This filters strongly negative tweets with high confidence.

```
print("\n16 Highly Negative Tweets")
print(df[(df['airline_sentiment'] == 'negative') &
         (df['airline_sentiment_confidence'] > 0.9)].head())
```

Output:

```
16 Highly Negative Tweets
   tweet_id airline_sentiment  ... hour word_count
3  570301031407624196      negative  ...   11         17
4  570300817074462722      negative  ...   11         10
5  570300767074181121      negative  ...   11         23
17 570276917301137409      negative  ...    9         27
24 570256553502068736      negative  ...    8         26

[5 rows x 17 columns]
```

Interpretation:

Represents serious complaints.

Problem 17: Pivot table analysis

Explanation:

This creates a summary of sentiment per airline.

```
print("\n17 Pivot Table")
pivot = pd.pivot_table(df,
                        values='tweet_id',
                        index='airline',
                        columns='airline_sentiment',
                        aggfunc='count')
print(pivot)
```

Output:

```
17 Pivot Table
airline_sentiment  negative  neutral  positive
airline
American           1960       463       336
Delta               955       723       544
Southwest          1186       664       570
US Airways          2263       381       269
United             2633       697       492
Virgin America      181        171       152
```

Interpretation:

Provides quick comparison.

Problem 18: Duplicate tweets

Explanation:

This detects duplicate tweet entries.

```
print("\n18 Duplicate Tweets")
print(df[df.duplicated(subset='text')].head())
```

Output:

```
18 Duplicate Tweets
      tweet_id airline_sentiment  ... hour word_count
331  568605449659895808      positive  ...    18         2
515   570299889688702976      positive  ...    11         2
1477 569705563287896064      positive  ...    19         3
1864 569545805826166784      positive  ...     9         3
1938 569513703722393601      positive  ...     7         3

[5 rows x 17 columns]
```

Interpretation:

Helps clean data.

Problem 19: Rank airlines

Explanation:

This ranks airlines based on positive sentiment.

```
print("\n19 Airline Ranking (Positive Sentiment)")
rank = df[df['airline_sentiment'] ==
'positive'].groupby('airline').size().rank(ascending=False)
print(rank)
```

Output:

```
19 Airline Ranking (Positive Sentiment)
airline
American      4.0
Delta         2.0
Southwest     1.0
US Airways    5.0
```

```
United          3.0
Virgin America  6.0
dtype: float64
```

Interpretation:

Shows best airlines.

Problem 20: Complex filtering

Explanation:

This applies multiple conditions to filter specific data.

```
print("\n20 United Negative Tweets")
print(df[(df['airline'] == 'United') &
         (df['airline_sentiment'] == 'negative')].head())
```

Output:

```
20 United Negative Tweets
      tweet_id airline_sentiment ... hour word_count
506  570307109704900608      negative ...    11         23
507  570307026263384064      negative ...    11         24
509  570306217001799680      negative ...    11         19
511  570304912468402177      negative ...    11         12
513  570302023993831425      negative ...    11         22

[5 rows x 17 columns]
```

Interpretation:

Useful for targeted analysis.

5. Conclusion

- Majority of tweets are negative
- Delay and service issues are common

- Some airlines perform better than others
- Data analysis helps improve services